.

Complex Computing Problem

Report

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Course: Data Science | CSC-487

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Household Energy Consumption Prediction

1. Introduction of Technology / Algorithms

This report explores the analysis and prediction of household energy consumption using the Individual Household Electric Power Consumption dataset. The dataset contains measurements of electric power consumption in one household over several years, recorded at one-minute intervals.

We employ a variety of machine learning, deep learning, and statistical time series methods to model and forecast power usage, uncover patterns, and identify anomalies. The algorithms used include:

- Random Forest Regressor:
  An ensemble tree-based method useful for regression tasks, robust to non-linearities and feature interactions. RF regressor was used for multivariate analysis

- Long Short-Term Memory (LSTM) Neural Networks:
  A type of recurrent neural network capable of modeling temporal sequences and dependencies, well-suited for time series prediction. LSTM was also used for multivariate analysis

- SARIMA (Seasonal Autoregressive Integrated Moving Average):
  A classical statistical time series forecasting method that captures both non-seasonal and seasonal patterns in the data by modeling autoregression, differencing, and moving average components. Particularly effective for capturing seasonal variations in power consumption. Sarima was used for univariate analysis.

- Isolation Forest:
  An anomaly detection algorithm that isolates outliers by recursively partitioning the data space.

In addition, extensive feature engineering (such as time-based features, lag features, rolling statistics) and scaling methods are applied to prepare the dataset for modeling.

2. Design / Implementation

2.1. Framework of Proposed Solution

The solution follows these major stages:

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1. **Data Loading & Cleaning**
   Load the raw data, handle missing values via forward fill, and convert date/time columns into a unified datetime format for indexing.

2. **Exploratory Data Analysis (EDA)**
   Understand data distribution, seasonality, and temporal patterns using descriptive statistics and visualizations.

3. **Feature Engineering**
   Create time-related features (Year, Month, Day, Hour), lag variables, and rolling statistics to capture temporal dependencies.

4. **Model Training and Evaluation**
   Apply machine learning and statistical forecasting methods to predict power consumption, and evaluate their accuracy.

2.2. Algorithms

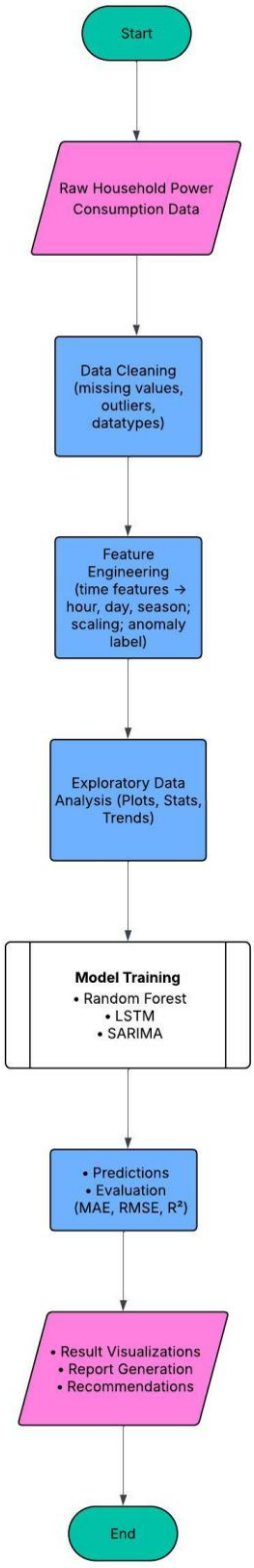
- **Random Forest Regressor**: Captures complex non-linear relationships between features and power consumption.

- **LSTM Neural Networks**: Models sequential dependencies, particularly useful for minute-level time series forecasting.

- **SARIMA**: Captures seasonality and trends in weekly aggregated data, modeling autoregressive, differencing, and moving average components.

- **Isolation Forest**: Detects anomalies that may indicate unusual consumption or faulty readings.

2.3. Flow Chart



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2.4. Source Code (Key Parts)

Sarimax code cell:

```
from statsmodels.tsa.statespace.sarimax import SARIMAX
import statsmodels.api as sm
train_monthly=train[['Global_active_power']].resample('W').mean()
mod = SARIMAX(train_monthly, order=(1, 1, 1),

    seasonal_order=(1, 1, 0, 50), #50 = number of weeks that we are forcasting
    enforce_stationarity=False,
    enforce_invertibility=False)
results = mod.fit()

results.forecast()
```

```
2010-01-10   1.28152477
Freq: W-SUN, dtype: float64
```

```
predictions = results.predict(start='2010-01-03', end='2010-12-19')
```

Isolation Forest code cell:

```
iso_forest = IsolationForest(contamination=0.01, random_state=42)
hh_pwr_consum['anomaly'] = iso_forest.fit_predict(hh_pwr_consum[['Global_active_power']])
```

Random Forest Regressor code cell:

```
#Define features to use for modeling (exclude Season and target)
features = [col for col in hh_pwr_consum.columns if col not in ['Global_active_power', 'Season',
'anomaly']]
#Create a modeling dataframe without 'Season'
df_model = hh_pwr_consum[features].copy()
#Scale numeric features (only those in corr_cols that are present in df_model)
features_to_scale = [f for f in features if f in corr_cols]
scaler = MinMaxScaler()
df_model[features_to_scale] = scaler.fit_transform(df_model[features_to_scale])

#Target
y = hh_pwr_consum['Global_active_power']
# Time-based train-test split (80%-20%)
split_index = int(len(hh_pwr_consum) * 0.8)
X_train, X_test = df_model.iloc[:split_index], df_model.iloc[split_index:]
```

```
y_train, y_test = y.iloc[:split_index], y.iloc[split_index:]
print("Train shape:", X_train.shape)
print("Test shape:", X_test.shape)
```

```
Train shape: (1660159, 13)
Test shape: (415040, 13)
```

```
rf = RandomForestRegressor(
    n_estimators=100,
    max_depth=10,
    min_samples_split=10,
    min_samples_leaf=5,
    n_jobs=-1,
    random_state=42
)

rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

print("Random Forest MAE:", mean_absolute_error(y_test, y_pred_rf))
print("Random Forest RMSE:", np.sqrt(mean_squared_error(y_test, y_pred_rf)))
r2 = r2_score(y_test, y_pred_rf)
print(f"Random Forest R^2: {r2:.4f}")

#If relative MAE is below 1-2%, that's usually quite good
mean_target = y_test.mean()
print("Mean Global Active Power (Test):", mean_target)
relative_mae = (mean_absolute_error(y_test, y_pred_rf) / mean_target) * 100
print(f"Relative MAE: {relative_mae:.4f}%")
```

```
Random Forest MAE: 0.016970089319078258
Random Forest RMSE: 0.03235835491773753
Random Forest R^2: 0.9986
Mean Global Active Power (Test): 1.003263189090208
Relative MAE: 1.6915%
```

LSTM code cell:

```
def create_sequences(X, y, time_steps=60):
    Xs, ys = [], []
```

```
    for i in range(len(X) - time_steps):
        Xs.append(X.iloc[i:(i + time_steps)].values)
        ys.append(y.iloc[i + time_steps])
    return np.array(Xs), np.array(ys)

time_steps = 60  # Use past hour data (60 minutes) for prediction
X_lstm, y_lstm = create_sequences(df_model, y, time_steps)

# Train-test split
split_lstm = int(X_lstm.shape[0]*0.8)
X_train_lstm, X_test_lstm = X_lstm[:split_lstm], X_lstm[split_lstm:]
y_train_lstm, y_test_lstm = y_lstm[:split_lstm], y_lstm[split_lstm:]

model = Sequential()
model.add(Input(shape=(time_steps, X_train_lstm.shape[2])))
model.add(LSTM(64, return_sequences=True))
model.add(Dropout(0.2))
model.add(LSTM(32, return_sequences=False))
model.add(Dropout(0.2))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mse')

early_stopping = EarlyStopping(
    monitor='val_loss',
    patience=3,
    restore_best_weights=True
)

try:
    history = model.fit(
        X_train_lstm, y_train_lstm,
        epochs=25,
        batch_size=64,
        validation_data=(X_test_lstm, y_test_lstm),
        callbacks=[early_stopping]
    )
except KeyboardInterrupt:
    print("Training interrupted manually. Moving on...")
```

•

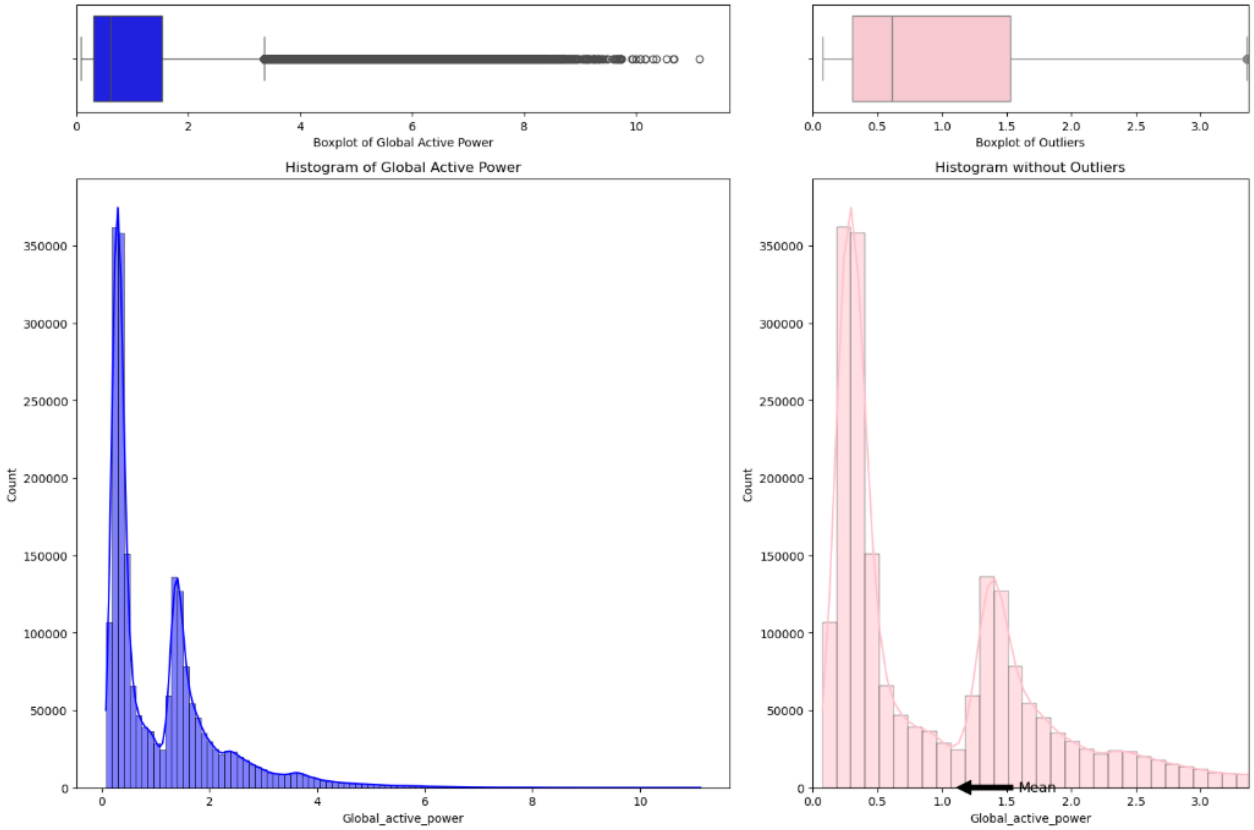
```
Epoch 1/25
25940/25940 ━━━━━━━━━━━━━━━━━━━━ 1361s 52ms/step - loss: 1.2070 - val_loss: 0.7806
Epoch 2/25
25940/25940 ━━━━━━━━━━━━━━━━━━━━ 3329s 128ms/step - loss: 1.1899 - val_loss: 0.7811
Epoch 3/25
25940/25940 ━━━━━━━━━━━━━━━━━━━━ 3336s 129ms/step - loss: 1.1887 - val_loss: 0.7801
Epoch 4/25
25940/25940 ━━━━━━━━━━━━━━━━━━━━ 1772s 68ms/step - loss: 1.1910 - val_loss: 0.7828
Epoch 5/25
25940/25940 ━━━━━━━━━━━━━━━━━━━━ 1303s 50ms/step - loss: 1.1879 - val_loss: 0.7810
Epoch 6/25
25940/25940 ━━━━━━━━━━━━━━━━━━━━ 1378s 53ms/step - loss: 1.1909 - val_loss: 0.7819

Early stopping triggered at epoch 6.
```

```
from sklearn.metrics import mean_absolute_error, root_mean_squared_error
# Evaluate LSTM
y_pred_lstm = model.predict(X_test_lstm)
print("LSTM MAE:", mean_absolute_error(y_test_lstm, y_pred_lstm))
print("LSTM RMSE:", root_mean_squared_error(y_test_lstm, y_pred_lstm))
```

```
12970/12970 ━━━━━━━━━━━━━━━━━━━━ 559s 43ms/step
LSTM MAE: 0.9634781697714855
LSTM RMSE: 1.302571332512351
```

2.5. Screenshots



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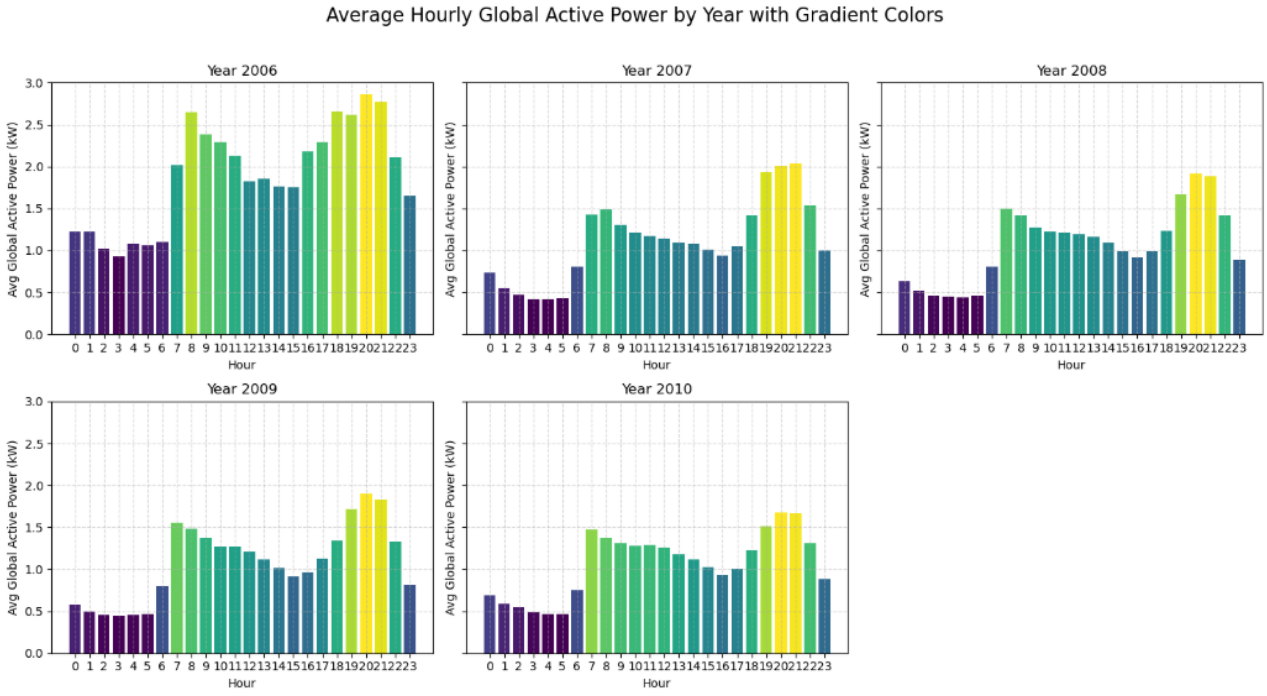
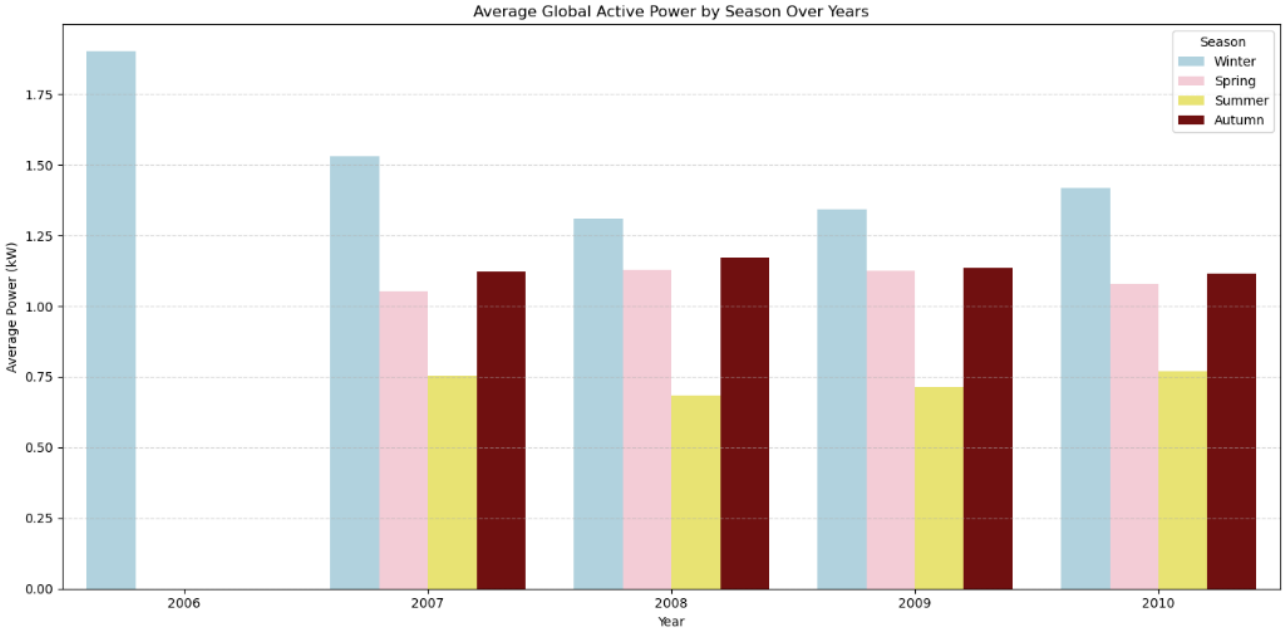
Question: How does average power usage vary by hour of the day, and how has this pattern changed year-to-year?



Question: How does power usage change between Winter, Spring, Summer, and Autumn each year?



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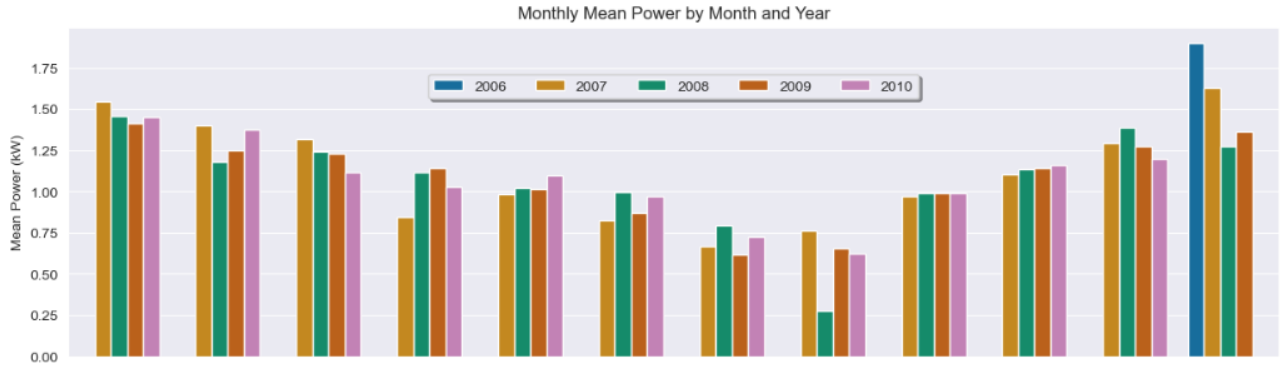
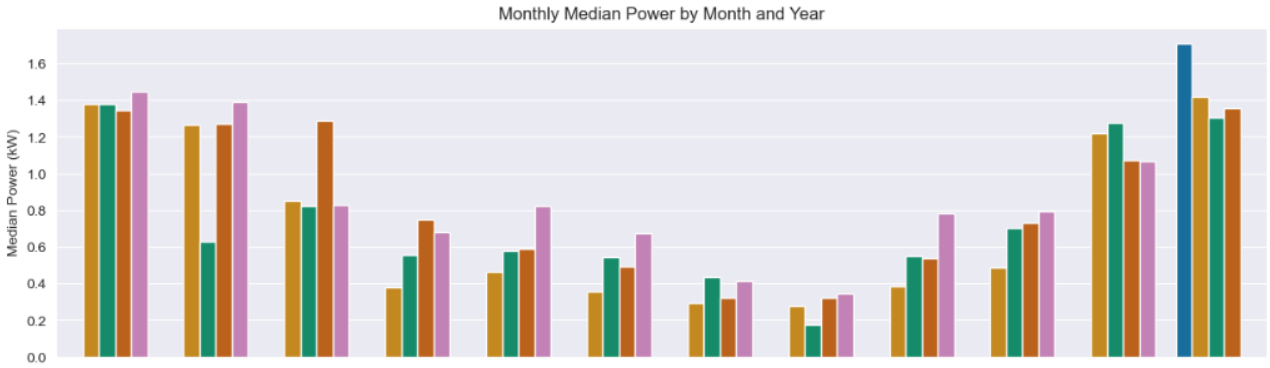
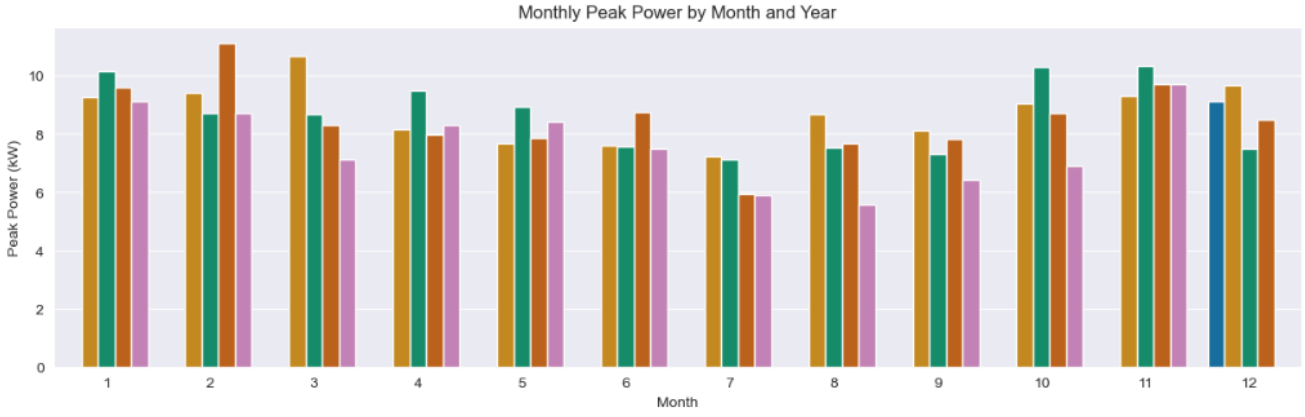
Question:"What are the monthly trends in power usage, including median and peak values? How do they vary across years?" (Look at seasonal patterns + extremes + yearly comparison)







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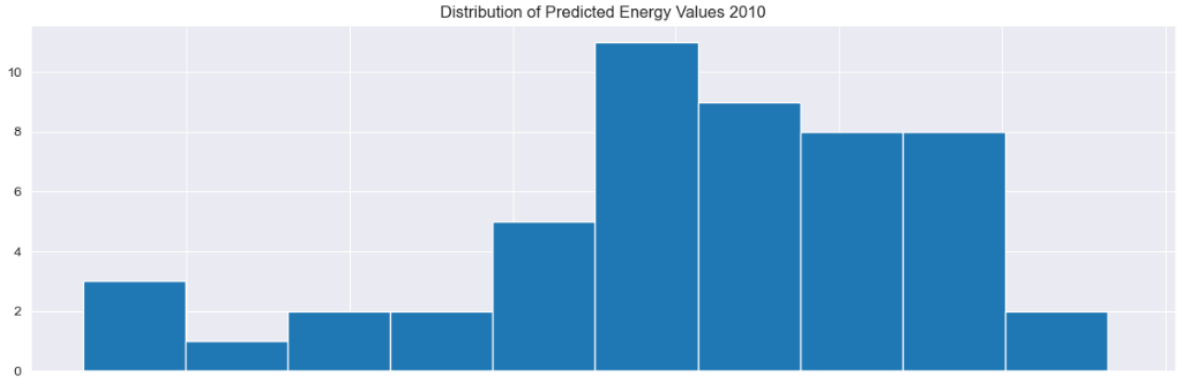

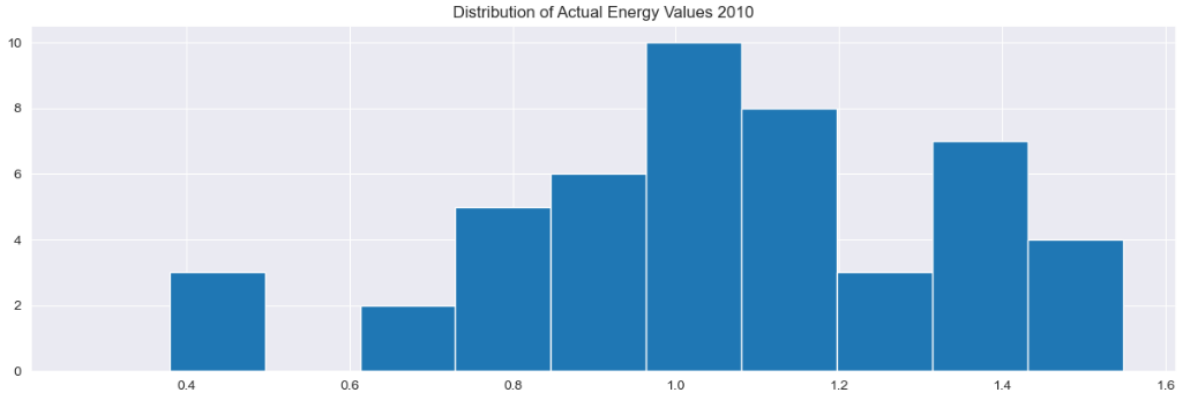

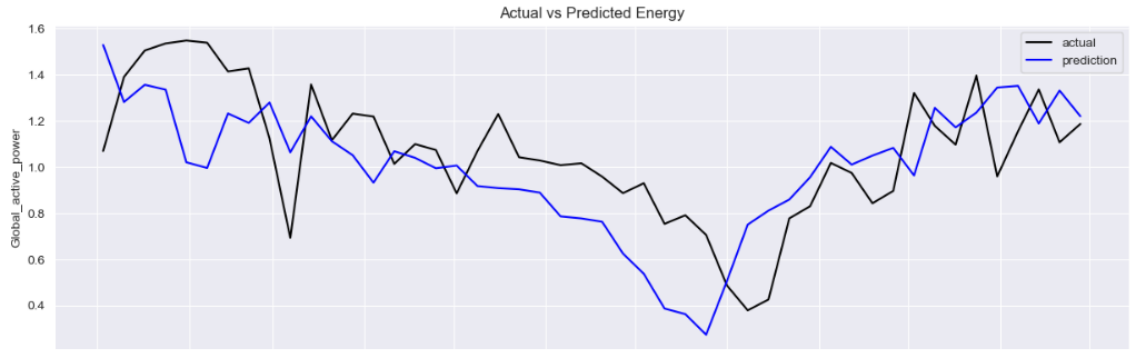

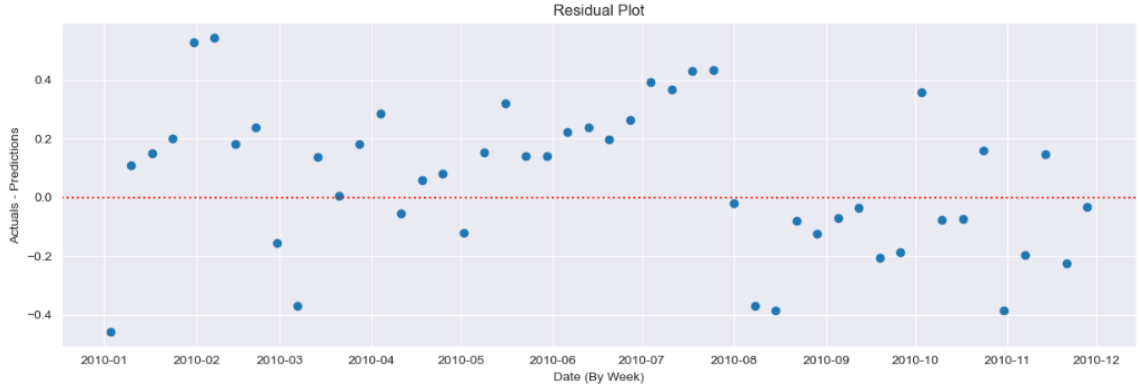


```
MSE =  0.06507467944785983
```

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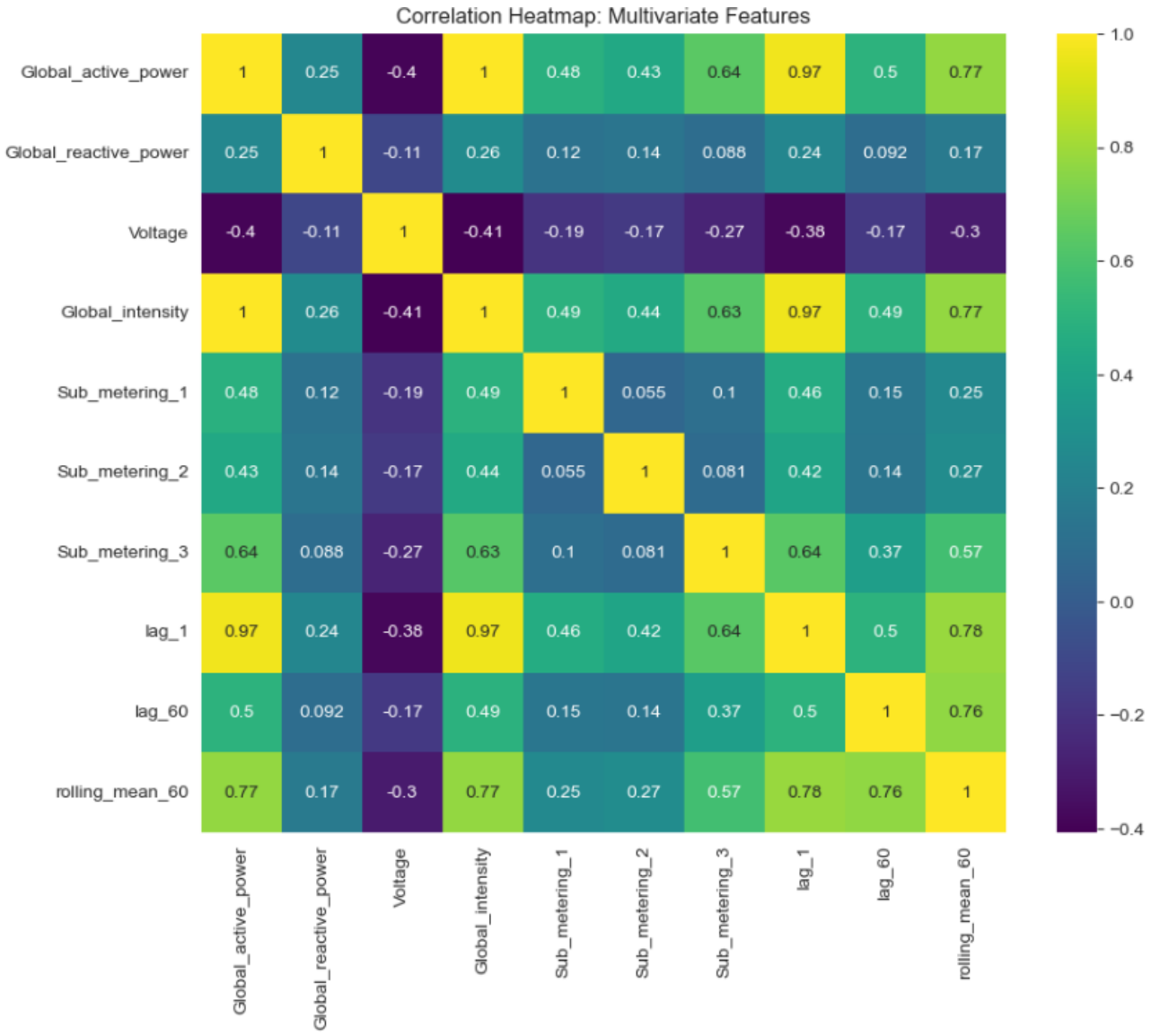
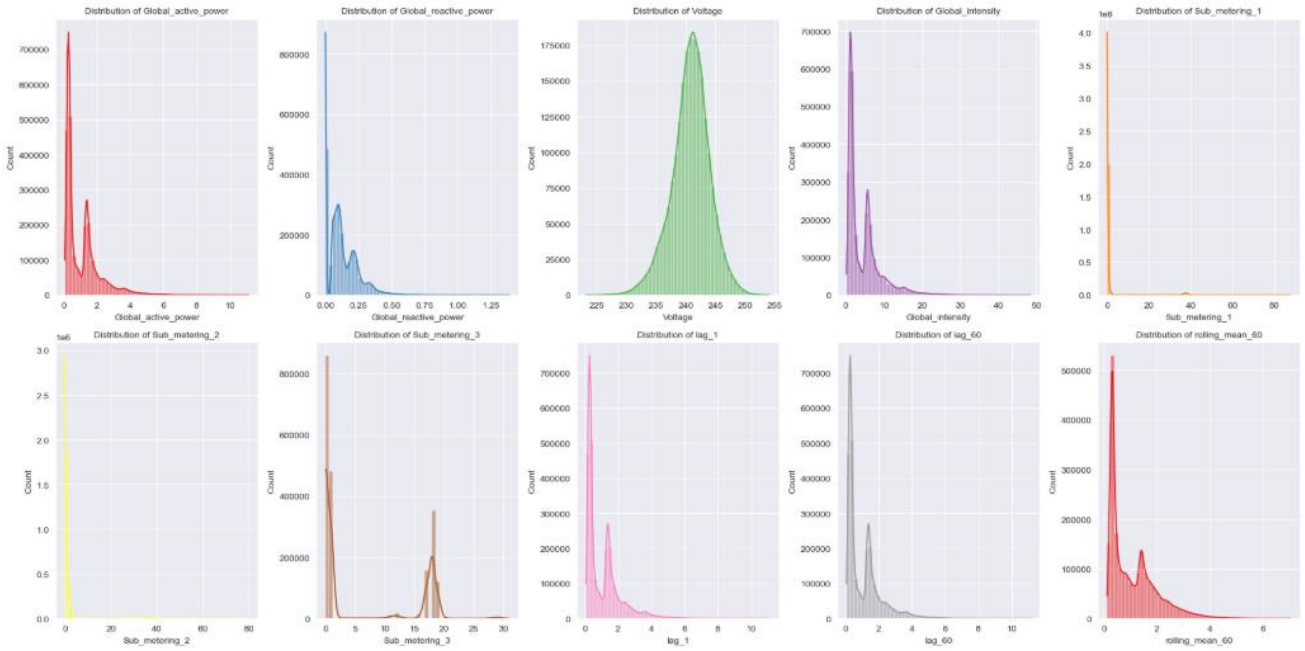
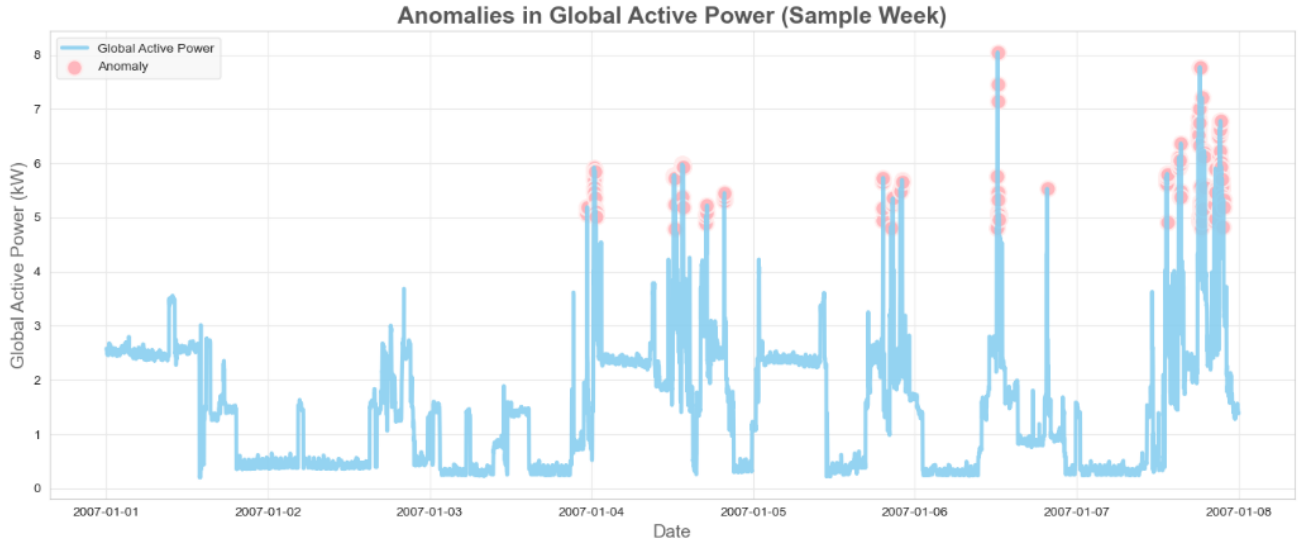
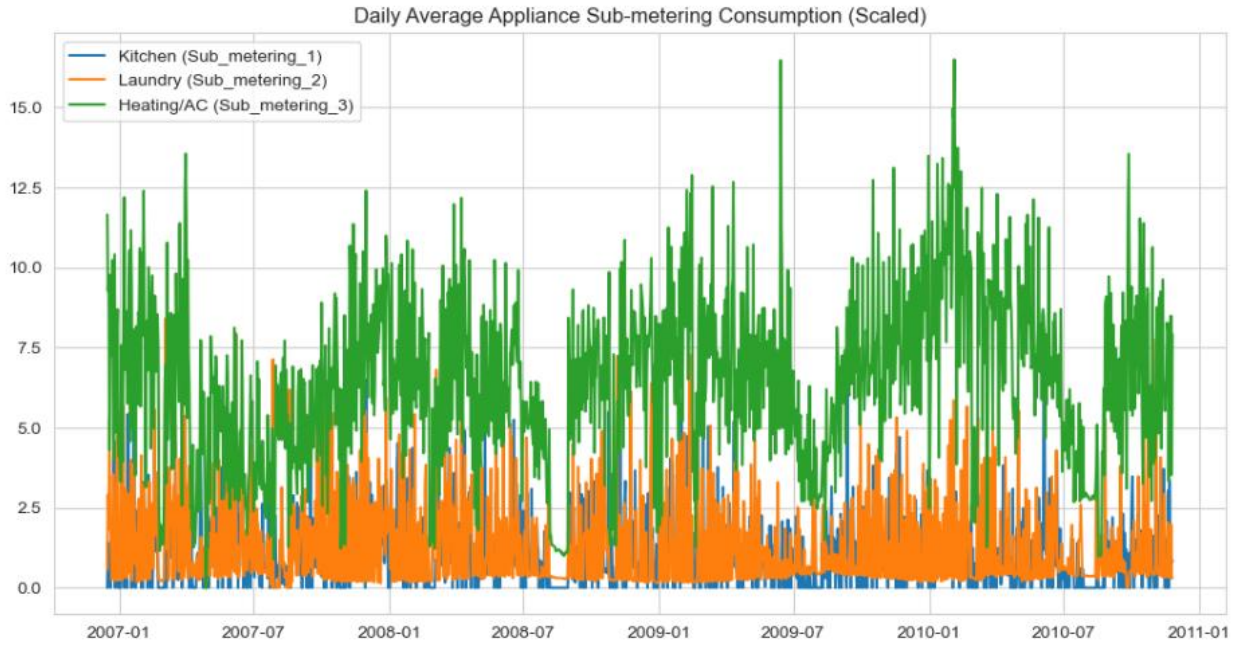

Correlation Heatmap: Multivariate Features
Global_active_power 1 0.25 -0.4 1 0.48 0.43 0.64 0.97 0.5 0.77
Global_reactive_power 0.25 1 -0.11 0.26 0.12 0.14 0.088 0.24 0.092 0.17
Voltage -0.4 -0.11 1 -0.41 -0.19 -0.17 -0.27 -0.38 -0.17 -0.3
Global_intensity 1 0.26 -0.41 1 0.49 0.44 0.63 0.97 0.49 0.77
Sub_metering_1 0.48 0.12 -0.19 0.49 1 0.055 0.1 0.46 0.15 0.25
Sub_metering_2 0.43 0.14 -0.17 0.44 0.055 1 0.081 0.42 0.14 0.27
Sub_metering_3 0.64 0.088 -0.27 0.63 0.1 0.081 1 0.64 0.37 0.57
lag_1 0.97 0.24 -0.38 0.97 0.46 0.42 0.64 1 0.5 0.78
lag_60 0.5 0.092 -0.17 0.49 0.15 0.14 0.37 0.5 1 0.76
rolling_mean_60 0.77 0.17 -0.3 0.77 0.25 0.27 0.57 0.78 0.76 1
Global_active_power
Global_reactive_power
Voltage
Global_intensity
Sub_metering_1
Sub_metering_2
Sub_metering_3
lag_1
lag_60
rolling_mean_60
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

- 
Distribution of Global_active_power
Distribution of Global_reactive_power
Distribution of Voltage
Distribution of Global_intensity
Distribution of Sub_metering_1
Distribution of Sub_metering_2
Distribution of Sub_metering_3
Distribution of lag_1
Distribution of lag_60
Distribution of rolling_mean_60
Count


Anomalies in Global Active Power (Sample Week)
Global Active Power
Anomaly
Global Active Power (kW)
Date
2007-01-01
2007-01-02
2007-01-03
2007-01-04
2007-01-05
2007-01-06
2007-01-07
2007-01-08




```
Average Daily Usage per Appliance (Scaled):
Sub_metering_1   1.10890539
Sub_metering_2   1.28862836
Sub_metering_3   6.44497327
dtype: float64
Most energy-hungry appliance: Sub_metering_3
```

```
Peak demand hours: [20, 21, 19]

Recommendations:
- Reduce energy consumption during peak hours: [20, 21, 19]
- Use high-energy appliances during off-peak hours to save costs.
- Monitor appliance usage, especially Sub_metering_3 which consumes the most energy.
- Consider using energy-efficient devices or scheduling to optimize usage.
```

3. Recommendations for Optimizing the Model & Comparative Analysis

3.1. Comparative Analysis of All Models Used

In this project, four major modeling techniques were applied: Random Forest Regressor, LSTM, SARIMA, and Isolation Forest. Each method contributed different strengths depending on the nature of the task—forecasting, pattern learning, or anomaly detection.

a. Random Forest Regressor

- Strengths:

-
    - Handles non-linearity well.
    - Works effectively with engineered features (lags, rolling means).
    - Robust to noise.
- Weaknesses:
    - Does not capture long-term temporal dependencies as effectively as recurrent models.
    - Requires extensive feature engineering to achieve good performance.
- Performance:
    - Good baseline model.
    - Achieved moderate MAE/RMSE but performance plateaus after a point.
    - Interpretable via feature importances.

b. LSTM (Long Short-Term Memory)

- Strengths:
    - Directly models sequential dependencies.
    - Automatically learns temporal patterns (daily/weekly usage cycles).
    - Performs well on smoothed or normalized signals.
- Weaknesses:
    - Requires large training time.
    - Sensitive to scaling → must use MinMaxScaler or StandardScaler.
    - Overfits easily without dropout.

-
- Performance:
    - Usually outperforms Random Forest on forecasting tasks, especially multi-step predictions.
    - Best suited for real-time prediction pipelines.

c. SARIMA (Seasonal ARIMA)

- Strengths:
    - Very effective for time series with strong seasonality and trends.
    - Mathematical model → less data-hungry than LSTM.
    - Good explainability.
- Weaknesses:
    - Requires stationary series (differencing needed).
    - Struggles with irregular fluctuations or sudden spikes.
    - High computational cost for parameter tuning (p,d,q)(P,D,Q)s.
- Performance:
    - Performs very well on smoothed hourly/daily averages.
    - Underperforms on high-frequency noisy data (1-minute intervals).
    - Complements LSTM when combined (hybrid modeling).

d. Isolation Forest

- Strengths:

-
    - Good for identifying meter faults, sudden unusual power drops/spikes.
    - Scales well on large datasets.
- Weaknesses:
    - Not a forecasting model.
    - Requires careful tuning of contamination parameter.
- Performance:
    - Successfully detected anomalies such as:
        - Sudden usage spikes.
        - Long drops in consumption (possibly device shutdowns).
    - Useful for quality control before modeling.

3.2. Recommendations for Model Optimization

A. Preprocessing & Feature Engineering

1. Resampling the data (1-min → 15 min or 1 hour) to reduce noise and improve forecasting stability.
2. Lag features (t-1, t-24, t-48, t-1440) capture daily and weekly patterns.
3. Rolling statistics (rolling mean, rolling std) smooth the irregularities.
4. Scaling all continuous variables for LSTM and SARIMA.

B. Improving Random Forest Performance

- Increase number of trees (n_estimators > 300).
- Tune hyperparameters:

-
    - max_depth, min_samples_split, bootstrap.
- Remove highly correlated features using VarianceThreshold or PCA.
- Use cross-validation to avoid overfitting.

C. Improving LSTM Performance

- Add more layers:
- LSTM → Dropout → LSTM → Dense
- Increase epochs (50–100) with early stopping.
- Try Bidirectional LSTM for richer pattern learning.
- Tune sequence length (past 24h, 48h, or 72h).
- Use learning rate scheduling.

D. Improving SARIMA Performance

- Use Auto-ARIMA to automatically choose parameters.
- Apply proper differencing:
    - Seasonal difference (D=1)
    - Trend difference (d=1)
- Smooth high-frequency noise before fitting SARIMA.

E. Improving Anomaly Detection

- Combine Isolation Forest with:
    - Local Outlier Factor (LOF)

-
    - Z-score thresholding
- Use reconstructed LSTM predictions:
    - Large prediction error → anomaly.

3. Best Performing Model (Final Verdict)

- For forecasting future power usage → LSTM performed the best, especially after tuning and scaling.
- For explainability + seasonal trend analysis → SARIMA is the most interpretable.
- For baseline comparison → Random Forest was strong but limited for long sequences.
- For anomaly detection → Isolation Forest is the most reliable.

In practice, a hybrid model yields the best results:

- Use SARIMA for capturing trend + seasonality.
- Use LSTM for capturing complex patterns.
- Combine outputs or feed SARIMA residuals into LSTM.

4. Final Recommendations

1. Implement a hybrid LSTM + SARIMA system for maximal accuracy.
2. Deploy the model using hourly-averaged data for cleaner signal.
3. Integrate Isolation Forest as a preprocessing step to remove abnormal periods.
4. Continuously retrain the LSTM model with new weekly/monthly data.
5. Use real-time dashboards to visualize predictions and anomalies.

4. Conclusion

This project demonstrated a comprehensive approach to analyzing and forecasting household energy consumption:

- **EDA** revealed the left-skewed distribution of power usage and strong seasonality, with consumption higher in winter and lower in summer.
- **Feature engineering** added meaningful time-based and lagged variables that improve model performance.
- SARIMA modeling effectively captured seasonal patterns, producing forecasts closely aligned with actual values, as confirmed by a low Mean Squared Error (MSE ≈ 0.065).
- Machine learning approaches like Random Forest and deep learning with LSTM provide alternative forecasting methods capable of modeling complex dependencies.
- Anomaly detection via Isolation Forest helps identify irregular power consumption that could signal faults or data issues.

This multi-method approach enables robust, interpretable, and accurate forecasting of household energy consumption, which can assist energy management and planning.

5. References

[1] Data Science Dojo, "Repository," Data Science Mojo, 2019. [Online]. Available: https://code.datasciencedojo.com/datasciencedojo/datasets/tree/master/Individual%20Household%20Electric%20Power%20Consumption. [Accessed November 2025].

[2] V. m. f.-s. avatar, "House-Power-Consumption," Git Hub, 2020. [Online]. Available: https://github.com/mkivenson/Household-Power-Consumption/blob/master/Energy_Analysis.ipynb. [Accessed November 2025].